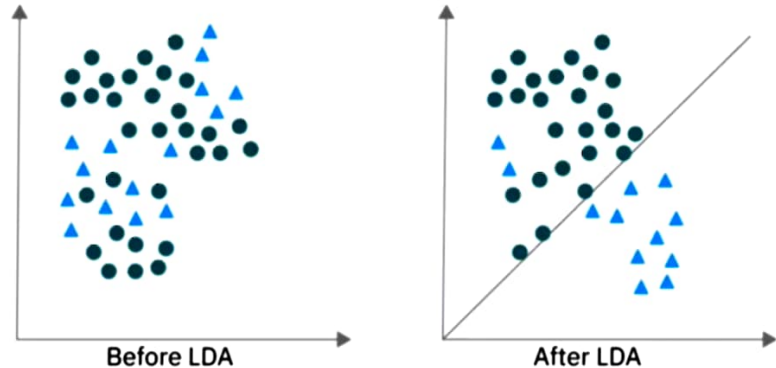


Linear Discriminant Analysis (LDA)

- ❑ Linear Discriminant Analysis (LDA) is a supervised machine learning technique used for classification and dimensionality reduction, aiming to find a linear combination of features that best separates multiple classes by maximizing class separability and minimizing within-class variance.



How the linear discriminant function works

- ❑ The linear discriminant function helps make decisions in classification problems by separating data points based on features and classifying them into different classes or categories. The computation process can be summarized in these key steps:

1. Calculate the between-class variance

The between-class variance is the separability between classes—the distance between the class means.

2. Calculate the within-class variance

The within-class variance is the distance between class means and samples.

3. Project the data into a lower-dimensional space

This maximizes the between-class variance and minimizes the within-class variance

How the linear discriminant function works

Mathematically, it tries to maximize:

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

Where:

- S_B : Between-class scatter matrix.
- S_W : Within-class scatter matrix.
- w : Projection vector.

Linear Discriminant Analysis (LDA)

Cell Size	Cell Shape	Tumour(Type)
5	3	Benign (0)
6	2	Benign (0)
4	4	Benign (0)
10	8	Malignant (1)
9	7	Malignant (1)
8	6	Malignant (1)

Step 1: Compute Class Means

Benign tumors (label 0):

Cell Size(Mean) $((5+6+4)/3)$
Cell Shape(Mean) $((3+2+4)/3)$ [5,3]

Malignant tumors (label 1):

Cell Size(Mean) $((10+9+8)/3)$
Cell Shape(Mean) $((8+7+6)/3)$ [9,7]

Linear Discriminant Analysis (LDA)

Cell Size	Cell Shape	Tumour(Type)
5	3	Benign (0)
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9	7	Malignant (1)
8	6	Malignant (1)

Step 2: Compute Scatter Matrices

For Benign:

[5,3] 5-5,3-3 [0,0]

[6,2] 6-5,2-3 [1,-1]

[4,4] 4-5,4-3 [-1,1]

For Malignant:

[10,8] 10-9,8-7 [1,1]

[9,7] 9-9,7-7 [0,0]

[8,6] 8-9,6-7 [-1,-1]

Linear Discriminant Analysis (LDA)

In LDA, we calculate the **within-class scatter matrix** S_W by adding the scatter matrices of **each class**.

For each class k :

$$S_k = \sum_{i=1}^{n_k} (x_i - \mu_k)(x_i - \mu_k)^T$$

Then sum across all classes:

$$S_W = \sum_{k=1}^K S_k$$

Linear Discriminant Analysis (LDA)

Covariance matrix (scatter):

$$\begin{aligned} S_{Benign} &= (1, -1)(1, -1)^T + (-1, 1)(-1, 1)^T \\ &= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \end{aligned}$$

First centered vector: $(0, 0)$:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Second centered vector: $(1, -1)$:

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \times \begin{bmatrix} 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Third centered vector: $(-1, 1)$:

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} \times \begin{bmatrix} -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

[5,3] 5-5,3-3 [0,0]

[6,2] 6-5,2-3 [1,-1]

[4,4] 4-5,4-3 [-1,1]

Linear Discriminant Analysis (LDA)

Scatter matrix:

For **Malignant**:

[10,8]	10-9,8-7	[1,1]
[9,7]	9-9,7-7	[0,0]
[8,6]	8-9,6-7	[-1,-1]

$$\begin{aligned} S_{Malignant} &= (1, 1)(1, 1)^T + (-1, -1)(-1, -1)^T \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \end{aligned}$$

Total Within-Class Scatter Matrix:

$$\begin{aligned} S_W &= S_{Benign} + S_{Malignant} \\ &= \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \end{aligned}$$

Linear Discriminant Analysis (LDA)

Step 3: Compute the Projection Vector w

$$\begin{aligned} w &= S_W^{-1}(\mu_{Benign} - \mu_{Malignant}) \\ &= \left(\frac{1}{4} \times I\right) ([5, 3] - [9, 7]) = \frac{1}{4}[-4, -4] = [-1, -1] \end{aligned}$$

Step 4: Project Data Onto the New Axis

$$z = w^T x$$

Linear Discriminant Analysis (LDA)

Benign Samples:

(5,3)	$(-1*5 + -1*3)$	-8
(6,2)	$(-1*6 + -1*2)$	-8
(4,4)	$(-1*4 + -1*4)$	-8

Malignant Samples:

(10,8)	$(-1*10 + -1*8)$	-18
(9,7)	$(-1*9 + -1*7)$	-16
(8,6)	$(-1*8 + -1*6)$	-14

Step 5: Classification Decision Boundary

A threshold around -12 would easily separate the two classes.

Linear Discriminant Analysis (LDA)

Step 5: Classification Decision Boundary

A threshold around -12 would easily separate the two classes.

- You compute the mean of projected values for Benign samples:

$$z_{Benign} = w^T \mu_{Benign}$$

- You compute the same for Malignant:

$$z_{Malignant} = w^T \mu_{Malignant}$$

Benign Samples:

(5,3)	$(-1*5 + -1*3)$	-8
(6,2)	$(-1*6 + -1*2)$	-8
(4,4)	$(-1*4 + -1*4)$	-8

Malignant Samples:

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Then the threshold is the midpoint between them:

$$\text{threshold} = \frac{z_{Benign} + z_{Malignant}}{2}$$

Linear Discriminant Analysis (LDA)

Step 5: Classification Decision Boundary

A threshold around -12 would easily separate the two classes.

- $w = [-1, -1]$
- $\mu_{Benign} = [5, 3]$
- $\mu_{Malignant} = [9, 7]$

Now compute:

$$z_{Benign} = w^T \mu_{Benign} = [-1, -1] \cdot [5, 3] = -5 - 3 = -8$$

$$z_{Malignant} = w^T \mu_{Malignant} = [-1, -1] \cdot [9, 7] = -9 - 7 = -16$$

Then:

$$\text{threshold} = \frac{-8 + (-16)}{2} = \frac{-24}{2} = -12$$